

AI FAQ Assistant Engine

Enterprise REST API for Semantic Query Processing, Knowledge Indexing & Real-time Support

1. Platform Executive Overview

The **AI FAQ Assistant API** enables enterprise applications to integrate dynamic, multi-modal knowledge search and conversational intelligence. Powered by hybrid retrieval-augmented generation (RAG) architecture, the system combines dense vector retrieval with structured BM25 lexical search to deliver accurate answers from proprietary documentation.

Low Latency

<120ms P95 semantic search and inference execution globally.

Multi-Tenant

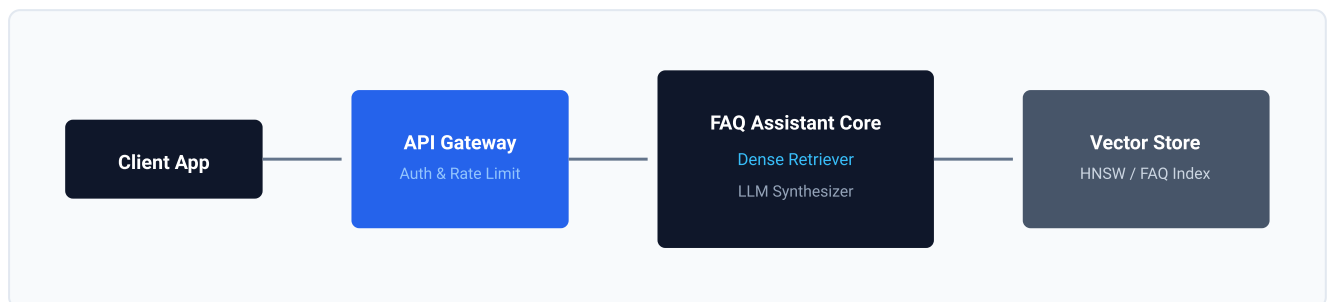
Isolated knowledge namespaces per environment or account.

Context Aware

Multi-turn session tracking with dynamic relevance filtering.

2. Global System Architecture

Requests flow through an API gateway layer enforcing strict TLS 1.3, token authentication, and token-bucket rate limiting before querying the core neural index.



3. Authentication & Security Policy

All REST API requests require a bearer API key passed in the standard Authorization HTTP header. Keys are scoped per workspace.

Authentication Header Format

```
Authorization: Bearer faq_live_sk_89f3a1d9e420b98a72c1
```

- **Production Keys:** Prefix `faq_live_`. Grants write access to live indexes.
- **Sandbox Keys:** Prefix `faq_test_`. Safe for staging environments and unit test mocks.

4. Core Interaction Endpoints

The Query API allows systems to send user prompts and receive ranked FAQ answers, confidence scores, and source attributions.

POST /v2/faq/query

Executes a real-time semantic query against the designated knowledge base collection.

Request Format	JSON
Rate Limit	1,000 requests per minute per key
Response Time	~95ms average

Request Body Payload Example:

```
{
  "knowledge_base_id": "kb_enterprise_v1",
  "query": "How do I reset my organizational single sign-on configuration?",
  "top_k": 2,
  "confidence_threshold": 0.75,
  "user_context": {
    "role": "admin",
    "region": "US-East"
  }
}
```

Response Structure (200 OK):

```
{
  "status": "success",
  "query_id": "qry_99214820a1",
  "answers": [
    {
      "faq_id": "faq_sso_001",
      "question": "How to reset enterprise SSO integration?",
      "answer": "Navigate to Admin Console > Security > Identity Providers, then click 'Regenerate Metadata'.",
      "score": 0.942,
      "metadata": { "category": "Security", "last_updated": "2026-01-15" }
    }
  ],
  "fallback_triggered": false
}
```

5. Knowledge Base Indexing API

Ingest and manage FAQ entry sets programmatically to maintain synchronized knowledge centers.

POST /v2/faq/knowledge-base/documents

Adds or updates items in the vectorized search index.

Field Name	Type	Required	Description
knowledge_base_id	String	Yes	Target namespace ID.
documents	Array[Object]	Yes	Collection of FAQ items (max 500 per batch).
reindex_now	Boolean	No	Forces immediate HNSW index build (Default: true).

6. Conversational Session Management

Maintain multi-turn support context by using session-based query orchestration.

POST /v2/faq/sessions

Initializes a dynamic session state with persistent short-term memory.

```
{
  "session_id": "sess_8830192a",
  "knowledge_base_id": "kb_enterprise_v1",
  "initial_prompt": "I need help with billing",
  "ttl_seconds": 3600
}
```

GET /v2/faq/sessions/{session_id}/history

Retrieves all dialogue turns and resolution metrics for a given conversation thread.

7. Webhook Event Notifications

Configure HTTP endpoints to receive real-time notifications regarding low confidence queries and indexing updates.

Event Name	Trigger Event	Payload Summary
query.unresolved	Confidence falls below threshold	Query string, top match scores, session ID.
kb.indexing.completed	Batch vectorization finishes	Indexed count, duration, updated KB ID.
feedback.submitted	End-user marks answer helpful/unhelpful	FAQ ID, rating (1-5), user comment.

Webhook Signature Verification Example:

All incoming webhooks contain a signature header `X-FAQ-Signature` computed using HMAC-SHA256.

```
// Signature Calculation
string payload = request.Body;
string secret = "whsec_99a8b1c2d3e4f5";
string expectedSignature = HMAC_SHA256(payload, secret);
```

Retry Mechanism

Failed webhook delivery attempts are retried up to 5 times with exponential backoff intervals (10s, 60s, 5m, 15m, 1h).

8. SDK Integration Examples

Python (Official SDK)

Install via pip: `pip install ai-faq-assistant`

```
from ai_faq import FAQClient

client = FAQClient(api_key="faq_live_sk_89f3a1d9e420b98a72c1")

response = client.query(
    knowledge_base_id="kb_enterprise_v1",
    query="What are the storage limits for standard accounts?",
    confidence_threshold=0.80
)

for answer in response.answers:
    print(f"[{answer.score}] {answer.question} -> {answer.answer}")
```

Node.js / TypeScript Integration

Install via npm: `npm install @ai-faq/sdk`

```
import { FAQAssistant } from '@ai-faq/sdk';

const faq = new FAQAssistant({
  apiKey: process.env.FAQ_API_KEY,
  environment: 'production'
});

async function runQuery() {
  const result = await faq.query({
    knowledgeBaseId: 'kb_enterprise_v1',
    query: 'How to configure custom domains?',
    topK: 3
  });

  console.log(`Matched ${result.answers.length} entries.`);
}

runQuery();
```

cURL Direct HTTP Request

```
curl -X POST https://api.faqassistant.ai/v2/faq/query \
-H "Authorization: Bearer faq_live_sk_89f3a1d9e420b98a72c1" \
-H "Content-Type: application/json" \
-d '{
  "knowledge_base_id": "kb_enterprise_v1",
  "query": "Where is my invoice?",
  "top_k": 1
}'
```

9. Error Codes & Diagnostic Matrix

The API uses standard HTTP response codes accompanied by structured error details in the JSON response payload.

HTTP Code	Error Code ID	Resolution Strategy
400 Bad Request	INVALID_PAYLOAD	Validate JSON formatting and required request parameters.
401 Unauthorized	AUTHENTICATION_FAILED	Verify API Key validity and Authorization header formatting.
404 Not Found	KB_NOT_FOUND	Confirm target knowledge_base_id exists in target workspace.
429 Rate Limited	THROTTLE_LIMIT_EXCEEDED	Implement exponential backoff based on <code>Retry-After</code> response header.
500 Server Error	INTERNAL_INDEX_ERROR	Retry request; if persistent, contact platform support.

10. Operational SLA & Compliance

Service Availability

99.95% monthly uptime SLA backed by multi-region global failover instances.

Security & Data Isolation

SOC 2 Type II certified. Query data is never utilized for public LLM training.

Data Retention Policy

Interactive queries and responses are retained for 30 days in encrypted audit logs for telemetry and fine-tuning before automatic purge. Namespace vector embeddings remain stored securely until explicit API call deletion.

11. Developer Support & Ecosystem

For technical inquiries, bug reporting, or quota adjustments:

- Developer Portal:** <https://developer.faqassistant.ai>
- System Status Dashboard:** <https://status.faqassistant.ai>
- API Engineering Support:** support@faqassistant.ai